Student Registration Form Deployment on AWS EC2 using NGINX, PHP & MariaDB

Project Overview

This project demonstrates deployment of a Student Registration Form web application on an Amazon EC2 instance using Amazon Linux.

The application allows users to submit registration details through a web form, and the entered data is stored successfully in a MariaDB database.

This project was developed to gain practical knowledge in:

- Cloud deployment
- Linux server administration
- Web hosting
- Database integration
- Backend connectivity using PHP

Technologies Used

Technology	Purpose
AWS EC2	Cloud Virtual Server
Amazon Linux	Operating System
NGINX	Web Server
MariaDB 10.5	Database
PHP	Backend
HTML	Frontend

Project Architecture

```
User Browser
      ↓
NGINX Web Server
      ↓
PHP Backend
```

```
        ↓
MariaDB Database
```

Implementation Steps

Step 1: Create EC2 Instance

Created an Amazon Linux EC2 instance and connected using SSH.

```
ssh -i "first.pem" ec2-user@ec2-13-221-32-203.compute-1.amazonaws.com
```

Step 2: Install Required Packages

Installed:

- NGINX
- MariaDB
- PHP

Command:

```
sudo yum install nginx mariadb105-server php php -y
```

Step 3: Start and Enable Services

Start services:

```
sudo systemctl start nginx mariadb php-fpm
```

Enable services:

```
sudo systemctl enable nginx mariadb php-fpm
```

Check status:

```
sudo systemctl status nginx mariadb php-fpm
```

Step 4: Create Frontend (HTML)

Move to web directory:

```
cd /usr/share/nginx/html/
```

Create signup file:

```
sudo vim signup.html
```

Paste frontend code for Student Signup Form.

Features

- Name
- Email
- Website
- Comment
- Gender
- Submit Button

Step 5: Configure Database

Login to MariaDB:

```
sudo mysql
```

Set root password:

```
alter user root@localhost identified by 'root';
```

Create database:

```
CREATE DATABASE FCT;
```

Use database:

```
USE FCT;
```

Create table:

```
CREATE TABLE users (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(20),
    email VARCHAR(100),
    website VARCHAR(255),
    gender VARCHAR(6),
    comment VARCHAR(100)
);
```

Step 6: Configure Backend (PHP)

Move to web directory:

```
cd /usr/share/nginx/html/
```

Create backend file:

```
sudo vim submit.php
```

Paste PHP backend code to:

- Receive form data
- Connect database
- Insert data into database
- Display successful submission message

Step 7: Install PHP MySQL Connector

Install connector:

```bash
sudo yum install php8.5-mysqlnd.x86_64 -y
```

Restart services:

```bash
sudo systemctl restart nginx mariadb php-fpm
```

Step 8: Access Application

Open browser and hit EC2 public IP:

```
http://13.221.32.203/signup.html
```

Student Registration Form opens successfully.

Step 9: Verify Data Entry

Check inserted data:

```bash
sudo mysql -u root -p
```

Enter password:

```
root
```

Use database:

```sql
USE FCT;
```

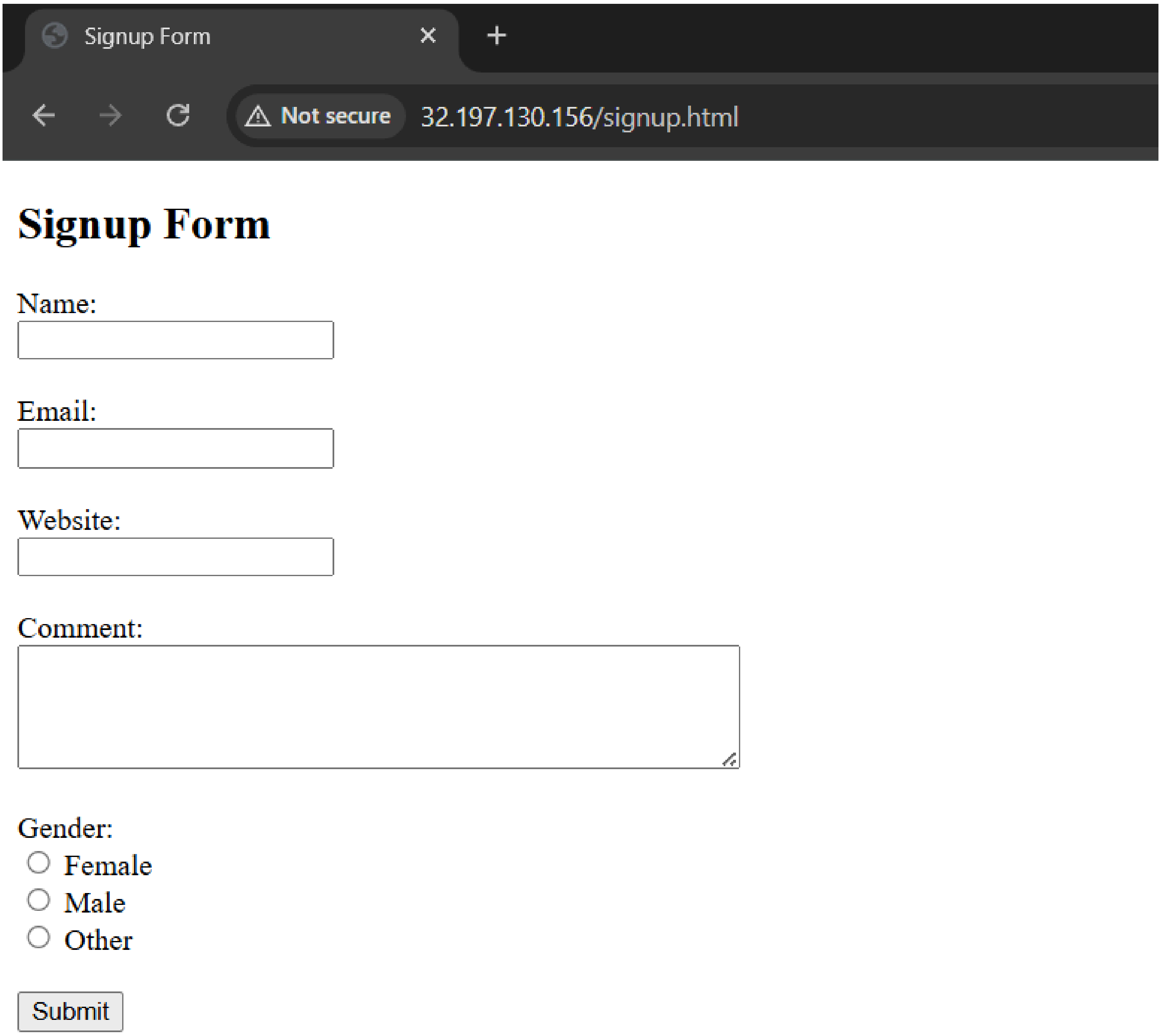
Display records:

```sql
SELECT * FROM USERS;
```

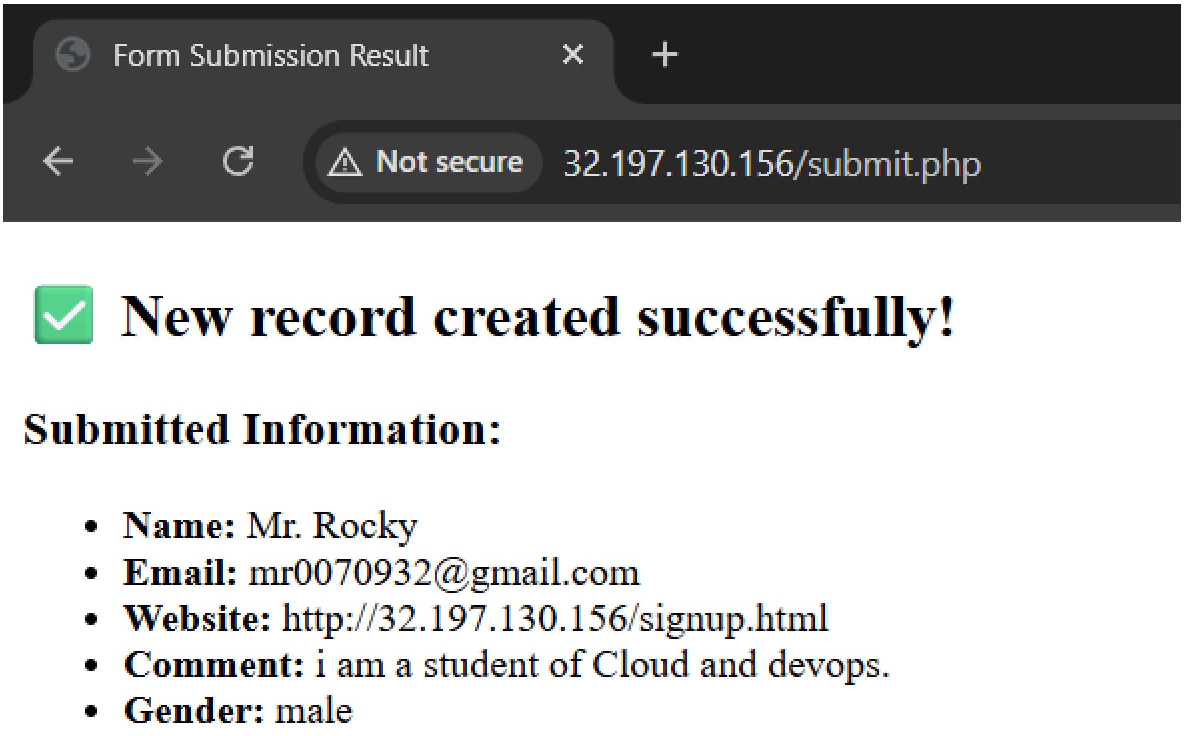
Data stored successfully in database.

Project Screenshots

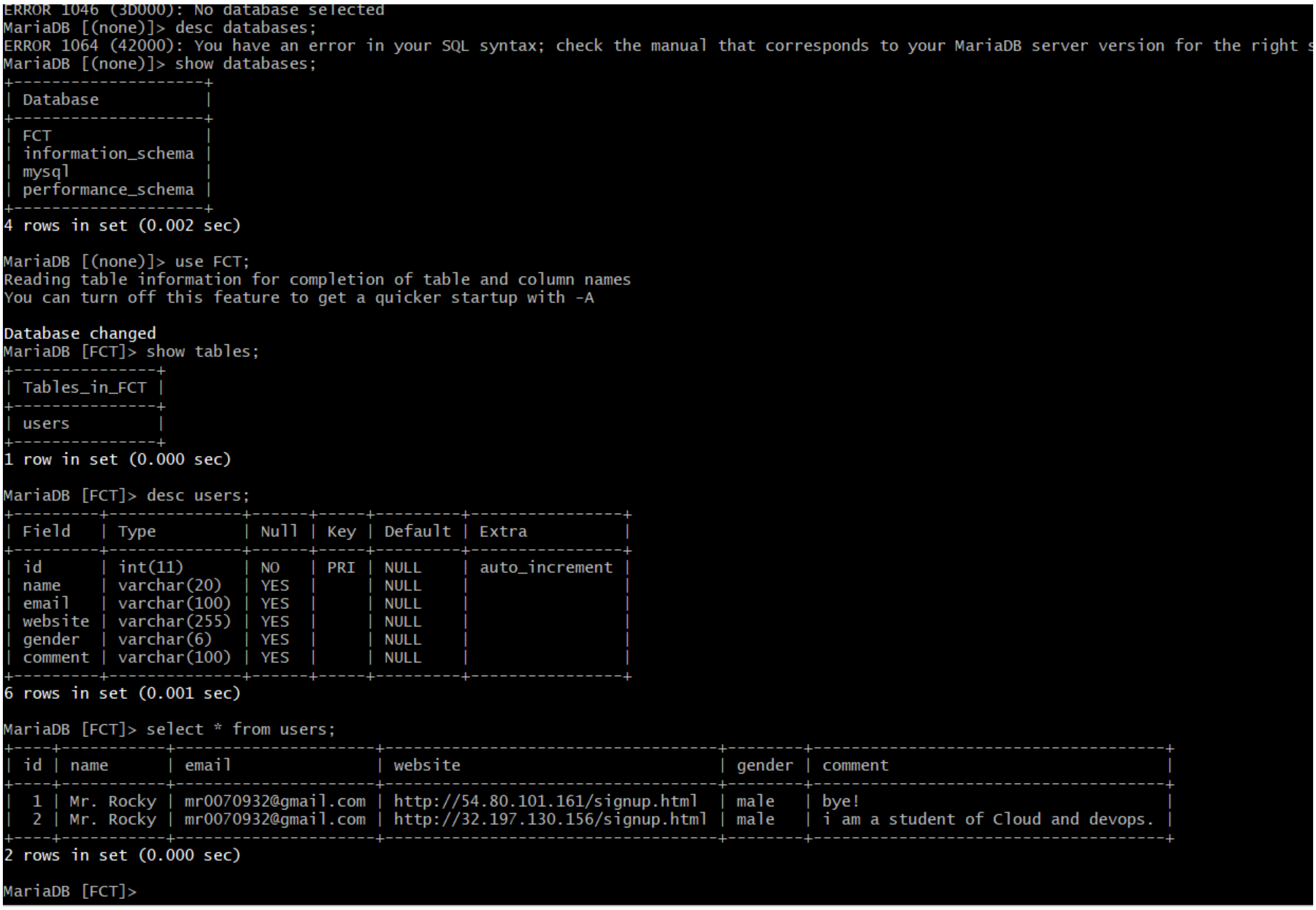
Registration Form



Form Submission Success



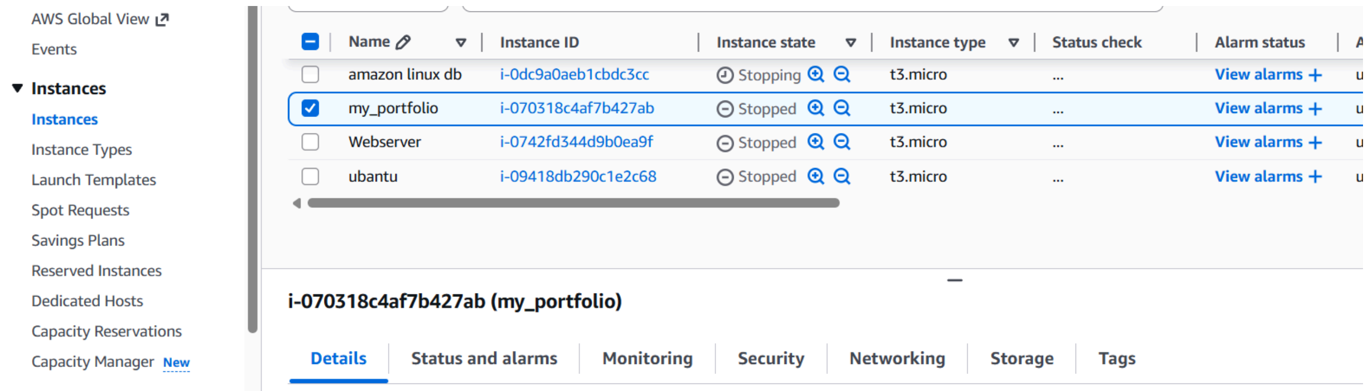
Database Records



```
ERROR 1046 (3D000): No database selected
MariaDB [(none)]> desc databases;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right s
MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| FCT                |
| information_schema |
| mysql              |
| performance_schema |
+--------------------+
4 rows in set (0.002 sec)

MariaDB [(none)]> use FCT;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [FCT]> show tables;
+---------------+
| Tables_in_FCT |
+---------------+
| users         |
+---------------+
1 row in set (0.000 sec)

MariaDB [FCT]> desc users;
+---------+--------------+------+-----+---------+----------------+
| Field   | Type         | Null | Key | Default | Extra          |
+---------+--------------+------+-----+---------+----------------+
| id      | int(11)      | NO   | PRI | NULL    | auto_increment |
| name    | varchar(20)  | YES  |     | NULL    |                |
| email   | varchar(100) | YES  |     | NULL    |                |
| website | varchar(255) | YES  |     | NULL    |                |
| gender  | varchar(6)   | YES  |     | NULL    |                |
| comment | varchar(100) | YES  |     | NULL    |                |
+---------+--------------+------+-----+---------+----------------+
6 rows in set (0.001 sec)

MariaDB [FCT]> select * from users;
+----+-----------+---------------------+-----------------------------------+--------+-----------------------------------------+
| id | name      | email               | website                           | gender | comment                                 |
+----+-----------+---------------------+-----------------------------------+--------+-----------------------------------------+
|  1 | Mr. Rocky | mr00709320gmail.com | http://54.80.101.161/signup.html  | male   | bye!                                    |
|  2 | Mr. Rocky | mr00709320gmail.com | http://32.197.130.156/signup.html | male   | i am a student of Cloud and devops.     |
+----+-----------+---------------------+-----------------------------------+--------+-----------------------------------------+
2 rows in set (0.000 sec)

MariaDB [FCT]>
```

Database Records

AWS EC2 Dashboard



Project Output

- Student registration form deployed successfully
- Form accessible using EC2 public IP
- User data stored in MariaDB database
- Full cloud deployment completed successfully

Learning Outcomes

Through this project I learned:

- AWS EC2 deployment
- Linux server management
- NGINX web server configuration
- MariaDB database setup
- PHP backend integration
- Full-stack cloud deployment
- Troubleshooting services

Problems Faced During Deployment

During deployment, some common issues were faced:

- Database connection errors
- PHP connector installation problems
- Service restart issues
- Linux command mistakes
- MariaDB syntax errors

These issues were solved using troubleshooting commands and service status checking.

Useful Commands

Show Databases

```
SHOW DATABASES;
```

Use Database

```
USE FCT;
```

Show Tables

```
SHOW TABLES;
```

Describe Table

```sql
DESC users;
```

Display Records

```sql
SELECT * FROM users;
```

Future Improvements

Future improvements planned for this project:

- Add CSS styling
- Add Bootstrap UI
- Add Login Authentication
- Add Password Encryption
- Add HTTPS SSL Certificate
- Deploy using Load Balancer
- Docker Containerization
- CI/CD Pipeline

Author

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Summary

This project is a simple cloud-based Student Registration Form application developed using AWS EC2, NGINX, PHP, and MariaDB.

The project provided practical experience in cloud hosting, Linux server management, backend integration, and database handling in a real-time environment.